- **Project Title**

  ESP32 RTC Digital Clock and Alarm

- **Objective**

  The main objective of this project is to design and develop a real-time digital clock using the ESP32 microcontroller and a DS3231 Real-Time Clock (RTC) module that displays the current date and time on a 16x2 LCD and includes a buzzer alarm feature

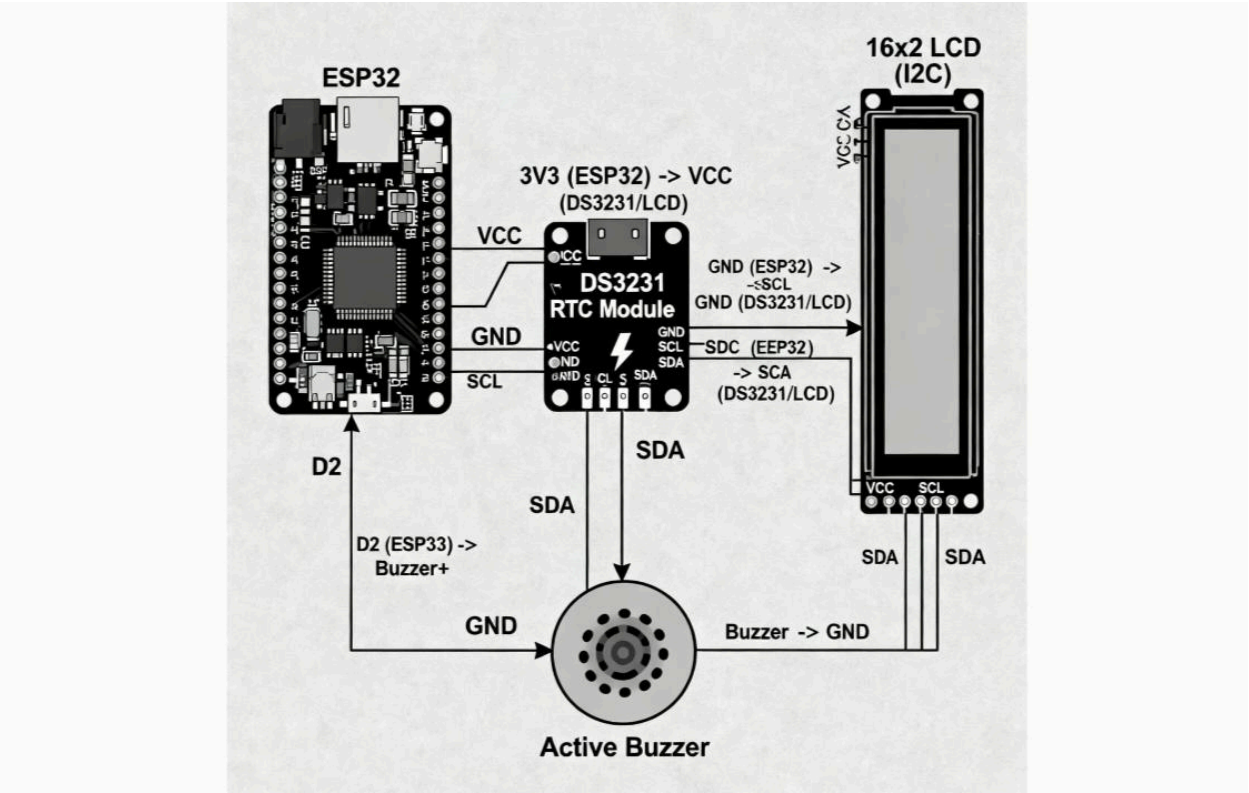
- **Components Required**

Components	Description
Esp32 Development Board	The main microcontroller used for time display and alarm control.
RTC Module DS3231	Maintains accurate date and time information.
LCD 16x2 with I2C Backpack	Displays real-time data (date and time).
Buzzer (Active / Passive)	Used for alarm indication.
Breadboard & Jumper Wires	For making circuit connections.

- **Circuit Connections**

Esp32 Pin	Components Pin(s)	Description
3.3V / VCC	RTC VCC, LCD VCC	Power supply for both modules
GND	RTC GND, LCD GND, Buzzer GND	Common ground connection
GPIO 21 (SDA)	RTC SDA, LCD SDA	I2C Data line
GPIO 22 (SCL)	RTC SCL, LCD SCL	I2C Clock line
GPIO 27	Buzzer Signal (+)	Digital output pin for alarm

- **Circuit Diagram**



- **Program Code: ESP32 RTC Digital Clock and Alarm**

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include "RTClib.h"

// Initialize LCD with I2C address (0x27 or 0x3F depending on your module)
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Initialize RTC (DS3231)
RTC_DS3231 rtc;

// Define buzzer pin
#define BUZZER_PIN 27

// Set alarm time (change this as needed)
int alarmHour = 7;     // 7 AM
int alarmMinute = 0;   // 00 Minutes
bool alarmTriggered = false;

void setup() {
  Wire.begin();             // Initialize I2C
  lcd.init();               // Initialize LCD
  lcd.backlight();          // Turn on LCD backlight
  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  // Start RTC
  if (!rtc.begin()) {
    lcd.setCursor(0, 0);
    lcd.print("RTC not found!");
    while (1);
  }

  // Uncomment this line only the first time to set the RTC to compile time:
```

```
  // rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));

  lcd.setCursor(0, 0);
  lcd.print("RTC Clock Ready");
  delay(1500);
  lcd.clear();
}

void loop() {
  DateTime now = rtc.now();  // Get current date and time

  // Display time on LCD
  lcd.setCursor(0, 0);
  lcd.print("Time: ");
  if (now.hour() < 10) lcd.print("0");
  lcd.print(now.hour());
  lcd.print(":");
  if (now.minute() < 10) lcd.print("0");
  lcd.print(now.minute());
  lcd.print(":");
  if (now.second() < 10) lcd.print("0");
  lcd.print(now.second());

  // Display date on LCD
  lcd.setCursor(0, 1);
  lcd.print("Date: ");
  lcd.print(now.day());
  lcd.print("/");
  lcd.print(now.month());
  lcd.print("/");
  lcd.print(now.year());

  // Check alarm
  if (now.hour() == alarmHour && now.minute() == alarmMinute &&
!alarmTriggered) {
```

```
    digitalWrite(BUZZER_PIN, HIGH);  // Turn ON buzzer
    delay(5000);                     // Buzzer ON for 5 seconds
    digitalWrite(BUZZER_PIN, LOW);   // Turn OFF buzzer
    alarmTriggered = true;
  }

  // Reset alarm flag when minute changes
  if (now.minute() != alarmMinute) {
    alarmTriggered = false;
  }

  delay(1000);
}
```

- **Working Principle**

  The ESP32 communicates with the DS3231 RTC using the I2C protocol to obtain accurate date and time data.
  This information is displayed on a 16x2 LCD via the I2C interface.
  When the system time matches the preset alarm time, the buzzer connected to GPIO 27 activates to alert the user.
  The DS3231 RTC maintains time even when the main power is turned off, due to its onboard backup battery.

- **Applications**

1. Digital wall clocks
2. Smart alarms
3. IoT-based time management systems
4. Real-time monitoring projects

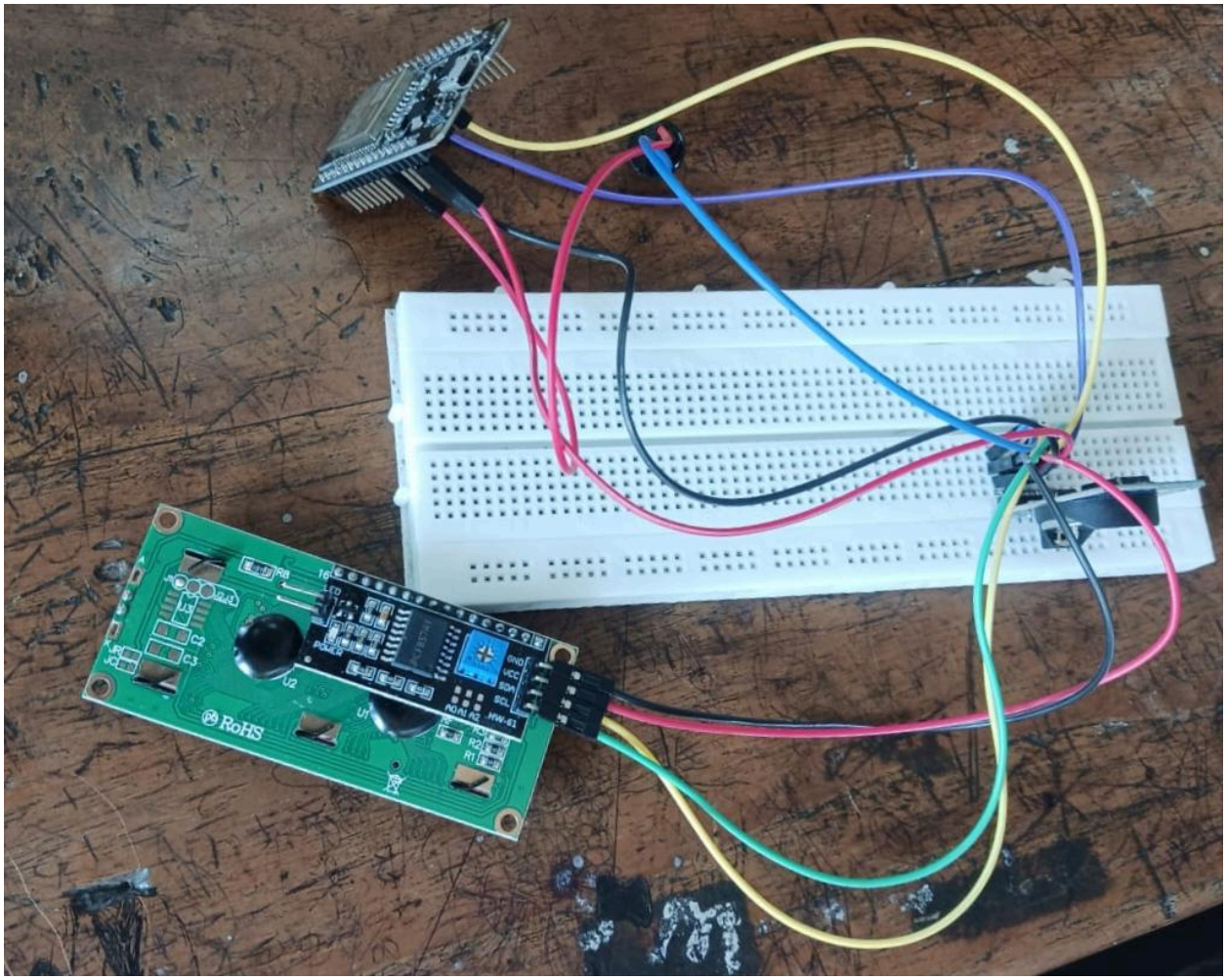
- **Advantages**

1. High accuracy using the DS3231 RTC module
2. Low power consumption
3. Easy to interface using I2C
4. Reliable and cost-effective design

- **Result**

  A real-time digital clock was successfully implemented using the ESP32 and DS3231 RTC module.
  The time and date are displayed on the LCD, and the buzzer alarm activates at the set time, demonstrating proper functionality of the system.

- **Conclusion**

  This project demonstrates the use of the ESP32 microcontroller for real-time applications using the DS3231 RTC module.
  It provides a practical understanding of I2C communication and basic alarm functionality, which can be expanded into advanced IoT-based smart time systems.

- **Project Photo**